
O-RAN Work Group 10 (OAM for O-RAN)

Topology Exposure and Inventory Data Model Specification - Stage 3

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Foreword

This Technical Specification (TS) has been produced by WG10 of the O-RAN ALLIANCE.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN ALLIANCE modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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- xx: the first digit-group is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc. (the initial approved document will have xx=01). Always 2 digits with leading zero if needed.
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Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

Executive summary

The present document specifies the TE&IV Data Models suited to realize the TE&IV Information Models as specified in [1].

1. Scope

This document specifies the TE&IV Data Models used to support TE&IV services within the SMO.

2. References

2.1. Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

- [1] O-RAN.WG10.TS.TE&IV-CIMI.0: "Topology Exposure & Inventory Common Information Models and Interface Specification - Stage 2" ("TE&IV CIMI")
- [2] O-RAN.WG10.TS.TE&IV-API.0: "Topology Exposure & Inventory Application Protocols Specification - Stage 3" ("TE&IV API")
- [3] IETF RFC 7951: "JSON Encoding of Data Modeled with YANG"

2.2. Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies. In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, O-RAN cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document, but they assist the user with regard to a particular subject area.

- [i.1] IETF RFC 7950: "The YANG 1.1 Data Modeling Language"

3. Definition of terms, symbols and abbreviations

3.1. Terms

Void

3.2. Symbols

Void

3.3. Abbreviations

Void

4. Solution Set (SS) definitions

4.1 YANG based Solution Set (SS) definitions

The present document defines the following TE&IV Solution Set definitions for O-RAN:

- YANG based ORAN TEIV Common YANG Types and Extensions Solution Set (Annex A).
- YANG based ORAN TE&IV RAN Domain Solution Set (Annex B).
- YANG based ORAN TE&IV Physical Domain Solution Set (Annex C).
- YANG based ORAN TE&IV REL-Physical-RAN Domain Solution Set (Annex D).
- YANG based User Defined Data Schema (Annex E).
- YANG based ORAN TE&IV O-Cloud Domain Solution Set (Annex F).

4.1.1 TE&IV Mapping from YANG to JSON schema

4.1.1.1 Introduction

As specified in Clause 5.3 of the TE&IV API Specification [2], responses with message content shall be in JSON format.

This clause defines the JSON mappings for the YANG schema specified in clause 4.1. Building on the principles outlined in RFC 7951 [3], the mappings specified here are extended to accommodate the YANG extensions in the O-RAN TE&IV YANG based Solution Set.

4.1.1.2 The “biDirectionalTopologyRelationship” Data Node

The biDirectionalTopologyRelationship data node describes a bi-directional relationship in the TE&IV model. It is a relationship comprising of an aSide and a bSide.

aSide and bSide are YANG extensions representing each end of the relationship. The order of aSide and bSide is important and must not be changed once defined. See TE&IV Common Information Models and Interface [1] Clause 4.2.1.1.4.1. Both aSide and bSide take a Topology Entity type as an argument. This entity type is represented by an instance identifier.

A biDirectionalTopologyRelationship instance is encoded as an object containing an array of nested objects, where each nested object represents child data nodes as its properties, as shown below.

A YANG example for the biDirectionalTopologyRelationship definition would be

```

or-teiv-yext:biDirectionalTopologyRelationship NFDEPLOYMENT_SERVES_ODUFUNCTION {
    ...
    key id;

    leaf-list served-oduFunction {
        description "O-DU Function served by this NF Deployment.";
        or-teiv-yext:aSide or-teiv-cloud:NFDeployment;
        type instance-identifier;
    }

    leaf-list serving-nFDeployment {
        description "NF Deployment that serves this O-DU Function.";
        or-teiv-yext:bSide or-teiv-ran:ODUFunction;
        type instance-identifier;
    }
}

```

the following is an example of a valid JSON-encoded instance:

```

{
  "o-ran-smo-teiv-rel-ran-cloud: NFDEPLOYMENT_SERVES_ODUFUNCTION": [
    {
      "id": "urn:<instance identifier of NFDeployment_serves_ODUFunction>",
      "serving-nFDeployment": "urn:<instance identifier of NFDeployment>",
      "served-oduFunction": "urn:<instance identifier of ODUFunction>",
      "sourceIds": [
        "urn:<instance source identifier of NFDeployment>",
        "urn:<instance source identifier of ODUFunction>"
      ]
    }
  ]
}

```

Annex A (normative): YANG definitions for common types and extensions

A.1 Generals

This annex contains the YANG definitions for the common types and extensions to the YANG language that topology and inventory model will frequently use, in accordance with information model definitions specified in TE&IV CIMI [1].

A.2 Common Modules

A.2.1 module o-ran-smo-teiv-common-yang-types

```
<CODE BEGINS>
module o-ran-smo-teiv-common-yang-types {
  yang-version 1.1;
  namespace "urn:o-ran-smo-teiv-common-yang-types";
  prefix or-teiv-types;

  import o-ran-smo-teiv-common-yang-extensions { prefix or-teiv-yext; }

  import _3gpp-common-yang-types { prefix types3gpp; }

  organization "O-RAN Alliance";

  contact
    "www.o-ran.org";

  description
    "Topology and Inventory common types model.
    This model contains re-usable data types that topology and inventory models
    will frequently use as part of types and relationships.

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    Licensed under the Standards Collaboration
    Copyright License Version 1.0 (the \"License\");
    You may not use this file except in compliance with the License.
    You may obtain a copy of the License at
    https://www.o-ran.org/sccl";

  revision "2025-11-19" {
    description "Replaced O-RAN Alliance License with SCCL License as per OSARP.";
    or-teiv-yext:label 1.2.1;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
  }

  revision "2025-06-30" {
    description "Added support for TE&IV User defined data";
    or-teiv-yext:label 1.2.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v03.00";
  }

  revision "2025-02-14" {
    description "Updated with GeoInformation.";
    or-teiv-yext:label 1.1.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
  }

  revision "2025-02-24" {
    description "Renamed Adapter Entity to Origin Entity. Added Geo-Information";
    or-teiv-yext:label 1.1.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
  }
}
```

```

revision "2024-10-10" {
    description "Initial revision.";
    or-teiv-yext:label 1.0.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v01.00";
}

grouping Top_Grp_Type {
    description "Grouping containing the key attribute common to all types. All types MUST use this
grouping.";

    leaf id {
        type string;
        description "Unique identifier of topology entities. Represents the Entity Instance Identifier.";
    }
}

grouping Origin_Entity_Mapping_Grp {
    description "Grouping to identify sourceIds on a topology object";

    leaf-list sourceIds {
        type sourceId;
        min-elements 1;
        ordered-by user;
        description
            "An ordered list of identities that represent the set of native
source identifiers for participating entities.
This must be attached to Topology Entity instance, outside of the
declared Topology Entity's attributes. This is exposed to the
consumers and can only be set by the responsible adapter. This
cannot be instantiated, and it MUST NOT be augmented or deviated
in any way, unless stated otherwise.";
    }
}

grouping Origin_Relationship_Mapping_Grp {
    description "Grouping to identify sourceIds on a topology relationship.";

    leaf-list sourceIds {
        type sourceId;
        min-elements 2;
        ordered-by user;
        description
            "An ordered list of identities that represent the set of native
source identifiers for participating entities in the relationship.
This must be attached to the Topology Relation instance, outside of the
declared Topology Relationship's attributes. This is exposed to the
consumers and can only be set by the responsible adapter. This
cannot be instantiated, and it MUST NOT be augmented or deviated in
any way, unless stated otherwise.";
    }
}

typedef sourceId {
    type string;
    description "An identity that represents a native identifier of a topology entity.";
}

grouping GeoInformation {
    description "A physical location with address and coordinates.";

    leaf address {
        type string;
        description "Address of the physical location.";
    }

    container coordinate {
        presence "Enable coordinate";
        uses types3gpp:GeoCoordinateGrp;
        description "Geographical coordinate containing Latitude, Longitude and Altitude. Reference:
3GPP TS 28.623";
    }
}

container decorators {
    description
        "This container serves as extension point for applications wishing

```

```

        to define their own decorators. This is done via augmentations. They
        can only be defined in name value pair.
        This is consumer data and can be attached to Topology Entity or
        Topology Relation instance, outside of the declared Topology Entity
        or Topology Relationship's attributes. This cannot be instantiated,
        and it MUST NOT be augmented or deviated in any way, unless stated
        otherwise.";
    }
    leaf-list classifiers {
        type identityref { base classifier; }
        description
            "Consumer defined tags to topology entities and relationships.
            This is consumer data and can be attached to Topology Entity or
            Topology Relation instance, outside of the declared Topology Entity
            or Topology Relationship's attributes. This cannot be instantiated,
            and it MUST NOT be augmented or deviated in any way, unless stated
            otherwise.";
    }

    identity classifier {
        description "The classifier is used as a base to provide all classifiers
            with identity. ";
    }
}
<CODE ENDS>

```

A.2.2 module o-ran-smo-teiv-common-yang-extensions

```

<CODE BEGINS>
module o-ran-smo-teiv-common-yang-extensions {
    yang-version 1.1;
    namespace "urn:o-ran-smo-teiv-common-yang-extensions";
    prefix or-teiv-yext;

    organization "O-RAN Alliance";

    contact
        "www.o-ran.org";

    description
        "Topology and Inventory YANG extensions model.
        This model contains extensions to the YANG language that topology and
        inventory models will use to define and annotate types and relationships.

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        Licensed under the Standards Collaboration
        Copyright License Version 1.0 (the \"License\");
        You may not use this file except in compliance with the License.
        You may obtain a copy of the License at
        https://www.o-ran.org/sccl";

    revision "2025-11-19" {
        description "Replaced O-RAN Alliance License with SCCL License as per OSARP.";
        or-teiv-yext:label 1.1.1;
        reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
    }

    revision "2025-02-14" {
        description "Updates for bi-directional relationship.";
        or-teiv-yext:label 1.1.0;
        reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
    }

    revision "2024-10-10" {
        description "Initial revision.";
        or-teiv-yext:label 1.0.0;
        reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v01.00";
    }
    extension biDirectionalTopologyRelationship {
        argument relationshipName;
        description
            "Defines a bi-directional relationship in the topology.
            A bi-directional-association (BDA) is a relationship comprising of
            an aSide and a bSide. The aSide is considered the originating
            side of the relationship; the bSide is considered the terminating

```

side of the relationship. The order of aSide and bSide is of importance and MUST NOT be changed once defined.

Both aSide and bSide are defined on a type, and are given a role. A type may have multiple originating and/or terminating sides of a relationship, all distinguished by role name.

The statement MUST only be a substatement of the 'module' statement. Multiple 'biDirectionalTopologyRelationship' statements are allowed per parent statement.

Substatements to the 'biDirectionalTopologyRelationship' define the aSide and the bSide, respectively, and optionally properties of the relationship. Data nodes of types 'leaf' and 'leaf-list' are used for this purpose. One of the data nodes MUST be annotated with the 'aSide' extension; another data node MUST be annotated with the 'bSide' extension. Other data nodes define properties of the relationship.

The argument is the name of the relationship. The relationship name is scoped to the namespace of the declaring module and MUST be unique within the scope.";

```
}
```

```
extension aSide {
  argument aSideType;
  description
    "Defines the aSide of a relationship.

    The statement MUST only be a substatement of a 'leaf' or 'leaf-list'
    statement, which itself must be a substatement of the
    'biDirectionalTopologyRelationship' statement.

    The data type of the parent 'leaf' or 'leaf-list' MUST be
    'instance-identifier'. Constraints MAY be used as part of the parent
    'leaf' or 'leaf-list' to enforce cardinality.

    The identifier of the parent 'leaf' or 'leaf-list' is used as name
    of the role of the aSide of the relationship. The name of the role
    is scoped to the type on which the aSide is defined and MUST be
    unique within the scope.

    While the parent 'leaf' or 'leaf-list' does not result in a property
    of the relationship, it is RECOMMENDED to avoid using the name of an
    existing type property as role name to avoid potential ambiguities
    between properties of a type, and roles of a relationship on the type.

    The argument is the name of the type on which the aSide resides.
    If the type is declared in another module, the type must be
    prefixed, and a corresponding 'import' statement be used to declare
    the prefix.";
```

```
}
```

```
extension bSide {
  argument bSideType;
  description
    "Defines the bSide of a relationship.

    The statement MUST only be a substatement of a 'leaf' or 'leaf-list'
    statement, which itself must be a substatement of the
    'biDirectionalTopologyRelationship' statement.

    The data type of the parent 'leaf' or 'leaf-list' MUST be
    'instance-identifier'. Constraints MAY be used as part of the parent
    'leaf' or 'leaf-list' to enforce cardinality.

    The identifier of the parent 'leaf' or 'leaf-list' is used as name
    of the role of the bSide of the relationship. The name of the role
    is scoped to the type on which the bSide is defined and MUST be
    unique within the scope.

    While the parent 'leaf' or 'leaf-list' does not result in a property
    of the relationship, it is RECOMMENDED to avoid using the name of an
    existing type property as role name to avoid potential ambiguities
    between properties of a type, and roles of a relationship on the type.
```

```

    The argument is the name of the type on which the bSide resides.
    If the type is declared in another module, the type must be
    prefixed, and a corresponding 'import' statement be used to declare
    the prefix.";
}

extension domain {
    argument domainName;
    description "Keyword used to carry domain information.";
}

extension label {
    argument semversion;
    description
        "The label can be used to give modules and submodules a semantic
        version, in addition to their revision.

        The format of the label is 'x.y.z' - expressed as pattern, it is
        [0-9]+\.\.[0-9]+\.\.[0-9]+

        The statement MUST only be a substatement of the revision statement.
        Zero or one revision label statements per parent statement are
        allowed.

        Revision labels MUST be unique amongst all revisions of a module or
        submodule.";
}
}
<CODE ENDS>
```

Annex B (normative): YANG definitions for O-RAN TE&IV RAN Domain

B.1 General

This annex contains the YANG definitions for the TE&IV RAN domain, in accordance with ORAN.SMO.TEIV.RAN information model definitions specified in TE&IV CIMI [1].

B.2 module o-ran-smo-teiv-ran

```
<CODE BEGINS>
module o-ran-smo-teiv-ran {
  yang-version 1.1;
  namespace "urn:o-ran:smo-teiv-ran";
  prefix or-teiv-ran;

  import o-ran-smo-teiv-common-yang-types {prefix or-teiv-types; }

  import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

  import _3gpp-common-yang-types { prefix types3gpp; }

  organization "O-RAN Alliance";

  contact
    "www.o-ran.org";

  description
    "RAN Logical topology model.
    This model contains the topology entities and relations in the
    RAN Logical domain, which represents the functional capability
    of the deployed RAN that are relevant to rApps use cases.

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    Licensed under the Standards Collaboration
    Copyright License Version 1.0 (the \"License\");
    You may not use this file except in compliance with the License.
    You may obtain a copy of the License at
    https://www.o-ran.org/sccl";

  revision "2025-11-19" {
    description "Replaced O-RAN Alliance License with SCCL License as per OSARP.";
    or-teiv-yext:label 1.1.1;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
  }

  revision "2025-02-14" {
    description "Revision to rename adapter group to origin group.";
    or-teiv-yext:label 1.1.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
  }

  revision "2024-10-10" {
    description "Initial revision.";
    or-teiv-yext:label 1.0.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v01.00";
  }

  or-teiv-yext:domain RAN;

  list ODUFunction {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "O-RAN Distributed Unit (O-DU).";
  }
}
```

The O-DU is an O-RAN NF in the O-RAN Architecture. An O-DU, combined with one or more O-RU(s) connected to it, supports and is fully compatible with the functions of a gNB-DU as defined by 3GPP TS 38.401.

The O-DU terminates the E2 and the F1 interface, and the Open Fronthaul interface (also known as LLS interface) as well as the RLC, MAC, and High-PHY functions of the radio interface towards the UE.

Note: O-DU is a concrete class that extends the ORANNetworkFunction abstract class. In Topology, you can create, read, update, and delete the O-DU object.

The management characteristics of the O-DU O-RAN NF are represented by the IOC GNBFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541";

```

container attributes {
  description "Container for O-RAN Distributed Unit (O-DU) attributes";
  leaf gNBID {
    type int64;
    description "Unique identifier for the DU within a gNodeB";
  }

  leaf gNBID {
    type int64;
    description "Identity of gNodeB within a PLMN";
  }

  leaf gNBIDLength {
    type int32;
    description "Length of gNBID bit string representation";
  }
}

```

```

list OCUCPFunction {
  uses or-teiv-types:Top_Grp_Type;
  uses or-teiv-types:Origin_Entity_Mapping_Grp;
  key id;
}

```

description "O-RAN Central Unit - Control Plane (O-CU-CP)

The O-CU-CP terminates the NG-c, X2-c, Xn-c, F1-c, and E1 interfaces as well as the RRC and PDCP (for SRB) protocols towards the UE as defined by 3GPP TS 38.401. The O-CU-CP terminates E2 interface to Near-RT RIC and is managed via O1 interface by the SMO.

Note: O-CU-CP is a concrete class that extends the ORANNetworkFunction abstract class. In Topology, you can create, read, update, and delete the O-CU-CP object.

The management characteristics of the O-CU-CP O-RAN NF are represented by the IOC GNBCUCPFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541";

```

container attributes {
  description "Container for O-RAN Central Unit - Control Plane (O-CU-CP) attributes";
  leaf gNBCUName {
    type string;
    description "Name of gNodeB-CU";
  }

  leaf gNBID {
    type int64;
    description "Identity of gNodeB within a PLMN";
  }

  leaf gNBIDLength {
    type int32;
    description "Length of gNBID bit string representation";
  }

  container pLMNID {
    presence "Enable pLMNID";
    description "PLMN identifier to be used as part of global RAN node identity";
    uses types3gpp:PLMNID;
  }
}

```

```

list OCUPFunction {
  uses or-teiv-types:Top_Grp_Type;
  uses or-teiv-types:Origin_Entity_Mapping_Grp;
  key id;
}

```

description "O-RAN Central Unit - User Plane (O-CU-UP)

The O-CU-UP terminates the NG-u, X2-u, Xn-u, F1-u, and E1 interfaces as well as the PDCP and SDAP protocols towards the UE as defined by 3GPP TS 38.401. The O-CU-UP terminates E2 interface to Near-RT RIC and is managed via O1 interface by the SMO.

Note: O-CU-UP is a concrete class that extends the ORANNetworkFunction abstract class. In Topology, you can create, read, update, and delete the O-CU-UP object.

The management characteristics of the O-CU-UP O-RAN NF are represented by the IOC GNBCUUPFunction as specified in 3GPP NR NRM model, as per 3GPP TS 28.541";

```

container attributes {
  description "Container for O-RAN Central Unit - User Plane (O-CU-UP) attributes";
  leaf gNBId {
    type int64;
    description "Identity of gNodeB within a PLMN";
  }

  leaf gNBIdLength {
    type int32;
    description "Length of gNBId bit string representation";
  }

  list pLMNIdList {
    key "mcc mnc";
    description "List of unique identities for PLMN";
    uses types3gpp:PLMNId;
  }
}

list NearRTRICFunction {
  uses or-teiv-types:Top_Grp_Type;
  uses or-teiv-types:Origin_Entity_Mapping_Grp;
  key id;

  description "Near-Real-Time RAN Intelligent Controller (Near-RT RIC)
    Near-RT RIC is an O-RAN NF that enables near real-time control and optimization of services and
    resources of E2 Nodes via fine-grained data collection and actions over the E2 interface with control
    loops in the order of 10 ms-1s.";

  container attributes {
    description "Container for Near-Real-Time RAN Intelligent Controller (Near-RT RIC) attributes";
    container pLMNId {
      description "PLMN identifier to be used as part of global RAN node identity";
      uses types3gpp:PLMNId;
    }

    leaf nearRtRicId {
      type int32;
      description "Identifier of Near-RT RIC";
    }
  }
}

list ORUFunction {
  uses or-teiv-types:Top_Grp_Type;
  uses or-teiv-types:Origin_Entity_Mapping_Grp;
  key id;

  description "O-RAN Radio Unit (O-RU)
    The O-RU terminates the Open Fronthaul interface (also known as LLS interface) as well as Low-PHY
    functions of the radio interface towards the UE. This is deployed as a PNF.
    The O-RU terminates the Open Fronthaul M-Plane interface towards the O-DU and SMO.";

  container attributes {
    description "Container for O-RAN Radio Unit (O-RU) attributes";
    leaf oruId {
      type string;
      description "Identity of the O-RU as discovered from the source domain based on M-Plane
architecture model";
    }
  }
}
}
<CODE ENDS>

```

Annex C (normative): YANG definitions for O-RAN TE&IV Physical Domain

C.1 General

This annex contains the YANG definitions for the TE&IV Physical domain, in accordance with ORAN.SMO.TEIV.Physical information model definitions specified in TE&IV CIMI [1].

C.2 module o-ran-smo-teiv-physical

```
<CODE BEGINS>
module o-ran-smo-teiv-physical {
  yang-version 1.1;
  namespace "urn:o-ran-smo-teiv-physical";
  prefix or-teiv-physical;

  import o-ran-smo-teiv-common-yang-types {prefix or-teiv-types; }

  import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

  organization "O-RAN Alliance";

  contact
    "www.o-ran.org";

  description
    "Physical domain topology model.
    This model contains the topology entities and relations in the
    Physical domain.

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    Licensed under the Standards Collaboration
    Copyright License Version 1.0 (the \"License\");
    You may not use this file except in compliance with the License.
    You may obtain a copy of the License at
    https://www.o-ran.org/sccl";

  revision "2025-11-19" {
    description "Replaced O-RAN Alliance License with SCCL License as per OSARP.";
    or-teiv-yext:label 1.1.1;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
  }

  revision "2025-06-30" {
    description "Updated revision to add container attribute for site.";
    or-teiv-yext:label 1.1.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v03.00";
  }

  revision "2025-02-10" {
    description "Initial revision.";
    or-teiv-yext:label 1.0.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
  }

  or-teiv-yext:domain PHYSICAL;

  list PhysicalAppliance {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;

    description "Represents the Physical Appliance in the Physical domain";

    container attributes {
      description "Container for Physical Appliance attributes";
      leaf vendorName {
```

```

        type string;
        description "Name of the physical appliance vendor";
    }

    leaf modelName {
        type string;
        description "Name of the physical appliance model";
    }
    leaf serialNumber {
        type string;
        description "Serial number of the physical appliance";
    }

    leaf applianceType {
        type string;
        description "Indicates the type of physical appliance";
    }
}

list Site {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the Site in the Physical domain";
    container attributes {
        description "Container for Site attributes";
        container siteLocation {
            description "Representation of physical site location";
            uses or-teiv-types:GeoInformation;
        }
        leaf siteName {
            type string;
            description "Human readable name of the physical site as identified by the mobile network operator";
        }
    }
}

or-teiv-yext:biDirectionalTopologyRelationship PHYSICALAPPLIANCE_INSTALLEDAT_SITE { // 1..n to 1..m
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Relationship_Mapping_Grp;
    key id;
    description "The aSide of this relationship is an instance of the PhysicalAppliance type.
    The bSide of this relationship is an instance of the Site type.";

    leaf-list installing-site {
        type instance-identifier;
        description "Site installing PhysicalAppliance.";
        or-teiv-yext:aSide PhysicalAppliance;
    }

    leaf-list installed-physicalAppliance {
        type instance-identifier;
        description "PhysicalAppliance installed at Site.";
        or-teiv-yext:bSide Site;
    }
}
}
<CODE ENDS>

```

Annex D (normative): YANG definitions for O-RAN TE&IV Physical-RAN Domain

D.1 General

This annex contains the YANG definitions for the TE&IV REL-Physical-RAN domain, in accordance with ORAN.SMO.TEIV.REL-Physical-RAN information model definitions specified in TE&IV CIMI [1].

D.2 module o-ran-smo-teiv-rel-physical-ran

```
<CODE BEGINS>
module o-ran-smo-teiv-rel-physical-ran {
  yang-version 1.1;
  namespace "urn:o-ran-smo-teiv-rel-physical-ran";
  prefix or-teiv-rel-phyran;

  import o-ran-smo-teiv-common-yang-types {prefix or-teiv-types; }

  import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

  organization "O-RAN Alliance";

  contact
    "www.o-ran.org";

  description
    "Physical to RAN Logical topology domain model.

    This model contains the topology relationship between the
    physical domain and the RAN domain.

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    Licensed under the Standards Collaboration
    Copyright License Version 1.0 (the \"License\");
    You may not use this file except in compliance with the License.
    You may obtain a copy of the License at
    https://www.o-ran.org/sccl";

  revision "2025-11-19" {
    description "Replaced O-RAN Alliance License with SCCL License as per OSARP.";
    or-teiv-yext:label 1.0.1;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
  }

  revision "2025-02-10" {
    description "Initial revision.";
    or-teiv-yext:label 1.0.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R004-v02.00";
  }

  or-teiv-yext:domain REL_PHYSICAL_RAN;

  or-teiv-yext:biDirectionalTopologyRelationship PHYSICALAPPLIANCE_SERVES_ODUFUNCTION { // 1..n to 1..m

    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Relationship_Mapping_Grp;
    key id;
    description "The aSide of this relationship is an instance of the PhysicalAppliance type.
    The bSide of this relationship is an instance of the ODUFunction type.";

    leaf-list served-oduFunction {
      type instance-identifier;
      description "ODUFunction served by PhysicalAppliance.";
      or-teiv-yext:aSide or-teiv-physical:PhysicalAppliance;
    }

    leaf-list serving-physicalAppliance {
```

```

        type instance-identifier;
        description "PhysicalAppliance serving ODUFfunction.";
        or-teiv-yext:bSide or-teiv-ran:ODUFfunction;
    }
}

or-teiv-yext:biDirectionalTopologyRelationship PHYSICALAPPLIANCE_SERVES_OCUCPFUNCTION { // 1..n to 1..m

    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Relationship_Mapping_Grp;
    key id;
    description "The aSide of this relationship is an instance of the PhysicalAppliance type.
    The bSide of this relationship is an instance of the OCUCPFfunction type.";

    leaf-list served-ocucpfFunction {
        type instance-identifier;
        description "OCUCPFfunction served by PhysicalAppliance.";
        or-teiv-yext:aSide or-teiv-physical:PhysicalAppliance;
    }

    leaf-list serving-physicalAppliance {
        type instance-identifier;
        description "PhysicalAppliance serving OCUCPFfunction.";
        or-teiv-yext:bSide or-teiv-ran:OCUCPFfunction;
    }
}

or-teiv-yext:biDirectionalTopologyRelationship PHYSICALAPPLIANCE_SERVES_OCUCUPFUNCTION { // 1..n to 1..m

    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Relationship_Mapping_Grp;
    key id;
    description "The aSide of this relationship is an instance of the PhysicalAppliance type.
    The bSide of this relationship is an instance of the OCUCUPfunction type.";

    leaf-list served-ocucupFunction {
        type instance-identifier;
        description "OCUCUPfunction served by PhysicalAppliance.";
        or-teiv-yext:aSide or-teiv-physical:PhysicalAppliance;
    }

    leaf-list serving-physicalAppliance {
        type instance-identifier;
        description "PhysicalAppliance serving OCUCUPfunction.";
        or-teiv-yext:bSide or-teiv-ran:OCUCUPfunction;
    }
}

or-teiv-yext:biDirectionalTopologyRelationship PHYSICALAPPLIANCE_SERVES_NEARRTRICFUNCTION { // 1..n to
1..m

    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Relationship_Mapping_Grp;
    key id;
    description "The aSide of this relationship is an instance of the PhysicalAppliance type.
    The bSide of this relationship is an instance of the NearRTRICfunction type.";

    leaf-list served-nearRTRICFunction {
        type instance-identifier;
        description "NearRTRICfunction served by PhysicalAppliance.";
        or-teiv-yext:aSide or-teiv-physical:PhysicalAppliance;
    }

    leaf-list serving-physicalAppliance {
        type instance-identifier;
        description "PhysicalAppliance serving NearRTRICfunction.";
        or-teiv-yext:bSide or-teiv-ran:NearRTRICfunction;
    }
}
}
<CODE ENDS>

```

Annex E (informative): YANG based User Defined Data Input Schema

E.1 General

This annex contains the YANG schema input files for the User Defined Data specified in TE&IV CIMI [1]. Before classifying the entities or the relationships in the topology network, the modules defined in the clauses below must be created and validated by using its own endpoint as shown in TE&IV API [2]. The user must provide a unique module name, to avoid collision of multiple users access that are defining classifiers and decorators. The schema cannot be modified later on but only deleted and recreated, if needed. Once a schema is successfully created and validated, the user can add the classifiers to the entities or relationships. The following clauses provide example templates for the schema input files.

E.2 Example Module (Decorators and Classifiers)

```
<CODE BEGINS>
module module-rapp-module {

    yang-version 1.1;
    namespace "urn:module-rapp-model";
    prefix module;

    import o-ran-smo-teiv-common-yang-types { prefix module; }
    import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

    revision "2024-06-10" {
        description
            "Initial revision.";
        or-teiv-yext:label 0.3.0;
    }

    augment /module:decorators {
        leaf decorator1 {
            type string;
        }
        leaf decorator2 {
            type boolean;
        }
        leaf decorator3 {
            type uint32;
        }
    }

    identity classifier1 {
        base module:classifiers;
    }

    identity classifier2 {
        base module:classifiers;
    }

    identity classifier3 {
        base module:classifiers;
    }
}
<CODE ENDS>
```

E.3 Example Module (Classifiers)

```
<CODE BEGINS>
module classifier-module-example {

    yang-version 1.1;
    namespace "urn:classifier-module-example";
    prefix module;

    import o-ran-smo-teiv-common-yang-types { prefix module; }
    import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

    revision "2025-05-10" {
        description
            "Initial revision.";
        or-teiv-yext:label 1.0.0;
    }

    identity classifier1 {
        base module:classifiers;
    }

    identity classifier2 {
        base module:classifiers;
    }

    identity classifier3 {
        base module:classifiers;
    }
}
<CODE ENDS>
```

E.4 Example Module (Decorators)

```
<CODE BEGINS>
module module-rapp-module {

    yang-version 1.1;
    namespace "urn:module-rapp-model";
    prefix module;

    import o-ran-smo-teiv-common-yang-types { prefix module; }
    import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }

    revision "2025-05-10" {
        description
            "Initial revision.";
        or-teiv-yext:label 0.3.0;
    }

    augment /module:decorators {
        leaf decorator1 {
            type string;
        }
        leaf decorator2 {
            type boolean;
        }
        leaf decorator3 {
            type uint32;
        }
    }
}
<CODE ENDS>
```

Annex F (normative): YANG definitions for O-RAN TE&IV O-Cloud Domain

F.1 General

This annex contains the YANG definitions for the TE&IV O-Cloud domain, in accordance with ORAN.SMO.TEIV.Cloud information model definitions specified in TE&IV CIMI [1].

F.2 module o-ran-smo-teiv-cloud

```
<CODE BEGINS>
module o-ran-smo-teiv-cloud {
  yang-version 1.1;
  namespace "urn:o-ran-smo-teiv-cloud";
  prefix or-teiv-cloud;
  import o-ran-smo-teiv-common-yang-types {prefix or-teiv-types; }
  import o-ran-smo-teiv-common-yang-extensions {prefix or-teiv-yext; }
  organization "O-RAN Alliance";
  contact
    "www.o-ran.org";
  description
    "Cloud domain topology model.
    This model contains the topology entities and relations in the
    Cloud domain.
    Copyright 2026 O-RAN ALLIANCE e.V.
    Licensed under the Standards Collaboration
    Copyright License Version 1.0 (the "License");
    you may not use this file except in compliance with the License.
    You may obtain a copy of the License at
    https://www.o-ran.org/sccl";

  revision "2025-10-23" {
    description "Initial revision.";
    or-teiv-yext:label 1.0.0;
    reference "O-RAN.WG10.TS.TE&IV-DM.0-R005-v04.00";
  }
  or-teiv-yext:domain CLOUD;

  list CloudifiedNF {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the Cloudified Network Function in the O-Cloud domain";
    container attributes {
      description "Container for CloudifiedNF attributes";
      leaf cloudifiedNfId {
        type string;
        description "Identifier of the CloudifiedNF";
      }
      leaf serviceOrchestrationId {
        type string;
        description "Identifier of the service orchestrator";
      }
    }
  }

  list NFDeployment {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the NF Deployment in the O-Cloud domain";
    container attributes {
      description "Container for NF Deployment attributes";
      leaf nfDeploymentId {
        type string;
        description "Identifier of the NF Deployment";
      }
    }
  }
}
```

```

        leaf resourceOrchestrationId {
            type string;
            description "Identifier of the resource orchestrator";
        }
        leaf serviceOrchestrationId {
            type string;
            description "Identifier of the service orchestrator";
        }
    }
}

list OCloudNamespace {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the O-Cloud Namespace in the O-Cloud domain";
    container attributes {
        description "Container for O-Cloud Namespace attributes";
        leaf oCloudNamespaceName {
            type string;
            description "Name of the O-Cloud Namespace";
        }
    }
}

list NodeCluster {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the Node Cluster in the O-Cloud domain";
    container attributes {
        description "Container for Node Cluster attributes";
        leaf nodeClusterId {
            type string;
            description "Identifier of the Node Cluster";
        }
        leaf nodeClusterName {
            type string;
            description "Name of the Node Cluster";
        }
        leaf resourceOrchestrationId {
            type string;
            description "Identifier of the resource orchestrator";
        }
        leaf serviceOrchestrationId {
            type string;
            description "Identifier of the service orchestrator";
        }
    }
}

list OCloudSite {
    uses or-teiv-types:Top_Grp_Type;
    uses or-teiv-types:Origin_Entity_Mapping_Grp;
    key id;
    description "Represents the O-Cloud Site in the O-Cloud domain";
    container attributes {
        description "Container for O-Cloud Site attributes";
        container oCloudSiteLocation {
            description "Representation of O-Cloud Site location";
            uses or-teiv-types:GeoInformation;
        }
        leaf oCloudSiteId {
            type string;
            description "Identifier of the O-Cloud Site";
        }
        leaf-list supportedMITs {
            type string;
            description "List of the supported MITs in the O-Cloud Site";
        }
    }
}

```

<CODE ENDS>

Annex (informative): Change History

Date	Revision	Description
2024.10.01	00.00.01	Initial proposed skeleton for the TE&IV Data Model Specification
2024.11.21	01.00	Implemented the following CRs for the November 2024 train: ERI-2024.08.14-WG10-CR-0128-Initial Content for the TE&IV Data Model Specification-v01, ERI-2024.10.09-WG10-CR-0130-TE&IV_RAN_Model-v02, ERI-2024.11.07-WG10-CR-0136-TE&IV_DM_ReferenceUpdate-v01
2025.02.07	02.00	Implemented the following CRs for the March 2025 train: ERI-2024.12.10-WG10-CR-0139-TE&IV Mapping from YANG to JSON schema-v02, ERI.AO-2025.02.10-WG10-CR-0152-Update to TE&IV YANG Schema to JSON output-v01, ERI-2025.02.25-WG10-CR-0154-Update to existing TE&IV yang modules, ERI-2025.02.25-WG10-CR-0155-Introducing new yang extensions, ERI.AO-2025.02.25-WG10-CR-0156-Yang module for the Rel-Physical-RAN namespace, ERI.AO-2025.02.25-WG10-CR-0157-Yang module for the Physical namespace
2025.07.01	03.00	Implemented the following CRs for the July 2025 train: ERI-2025.05.20-WG10-CR-0176-YANG-Definitions-for-Classifiers-and-Decorators-v02, ERI-2025.05.26-WG10-CR-0177-Updating-3GPP-reference-v01, ERI-2025.06.25-WG10-CR-0188-Data Model editorials-v02. Updated the specification for formatting corrections and editorial updates based on ODR and TS template v04
2025.11.19	04.00	Implemented the following CRs for the November 2025 train: ERI-2025.07.21-WG10-CR-0190-YANG Definitions Physical-v01, ERI-2025.09.16-WG10-CR-0199-Presence Field-v02, ERI-2025.10.27-WG10-CR-0210-Cloud YANG Data Model-v03, ERI-2025.11.06-WG10-CR-0211-Replacing O-RAN Alliance license with SCCL license in TE&IV YANG Modules-v01